ZZLIGHT PROJECT

ESP32 Wireless Art-Net to DMX512 Node

Laporan Pembuatan, Desain Hardware, Skematik & Dokumentasi Proyek DIY Open-Source

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Status Proyek: Production-Ready v4.0.0 (Final Stable Release)
Lisensi: Open Source MIT License

Hardware Core: ESP32 Dual-Core (FreeRTOS SMP) + MAX485
Protokol: Art-Net 4 (UDP 6454) & USITT DMX512-A
AI Research Session: Google Gemini Research #9b15b21a9066dd07

1. Ikhtisar Proyek & Hasil Akhir (Live Setup)

Proyek ini menghadirkan modul konverter **Wireless Art-Net to DMX512** mandiri (standalone) yang murah, berkinerja tinggi, dan bebas jitter untuk sistem pencahayaan panggung. Modul ini menerima data kontrol lighting nirkabel dari laptop melalui WiFi 2.4 GHz (Art-Net UDP 6454), lalu mengonversinya secara real-time menjadi sinyal elektrikal RS-485 DMX512 standar industri untuk menggerakkan lampu panggung fisik seperti PAR LED, Moving Head, Laser, dan Dimmer Pack tanpa kabel DMX panjang dari FOH.

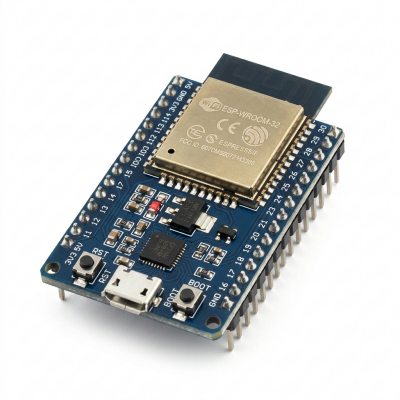
Gambar 1.1: Hasil Akhir Live Setup Modul ESP32 Art-Net DMX dalam Box X4 terhubung ke Laptop Acer Swift 3 (QLC+) dan mengontrol 4 lampu PAR LED 54 mata (Ungu, Merah, Oranye, Hijau)

2. Fitur Unggulan & Arsitektur Utama

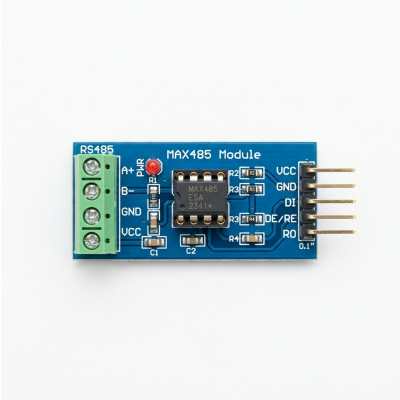
• **FreeRTOS Dual-Core SMP:** Core 0 khusus mengeksekusi transmisi DMX secara deterministik (44.1 Hz, Zero-Jitter), sedangkan Core 1 menangani WiFi Art-Net, WebServer, dan DNS.
• **Atomic Double-Buffering (Ping-Pong Memory):** Menghilangkan frame tearing dan kedipan lampu panggung saat ada data baru masuk.
• **Smart Captive Portal:** Otomatis membuka laman konfigurasi WiFi saat smartphone/PC terhubung ke hotspot ZZLIGHT-AP.
• **11-State Machine LCD 16x2:** Menampilkan status koneksi, Universe aktif, dan live FPS DMX secara real-time.
• **10-Detik Fail-Safe Blackout:** Otomatis memadamkan lampu jika software laptop terputus lebih dari 10 detik.

3. Daftar Alat dan Bahan (Bill of Materials)

Total biaya perakitan modul ini sangat terjangkau (< Rp 150.000 / < $10). Seluruh komponen dan alat yang digunakan terdokumentasi dalam katalog berikut:



Gambar 2.1
ESP32 DevKit V1 30-Pin

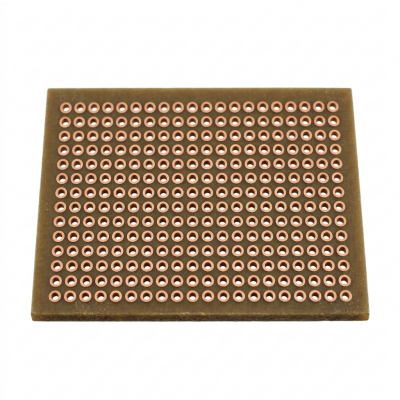


Gambar 2.2
Modul MAX485 TTL to RS-485

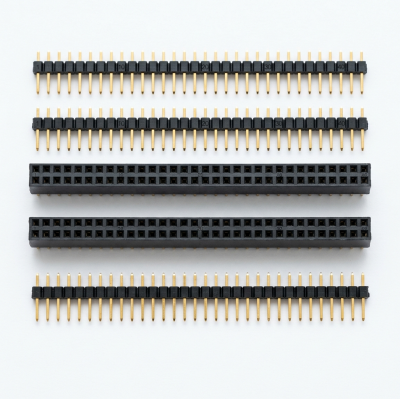
Gambar 2.3
LCD 16x2 + I2C PCF8574

Gambar 2.4
Soket Sasis XLR 3-Pin Female

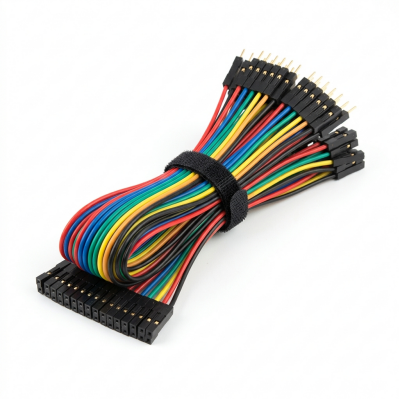
Gambar 2.5
Box Enclosure Plastik X4



Gambar 2.6
PCB Matrix Lubang (Rawboard)



Gambar 2.7
Pin Header Male & Female



Gambar 2.8
Kabel Jumper DuPont Multi-Warna

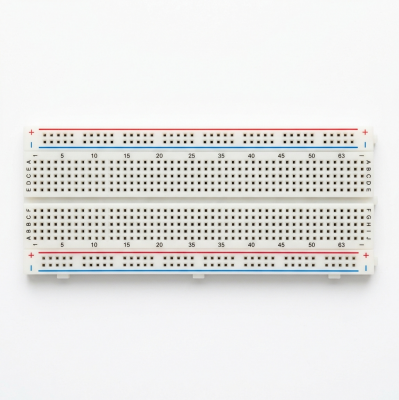
Gambar 2.9
Kabel USB ke Micro-USB Data

Gambar 2.10
Timah Solder 60/40 0.8mm

Gambar 2.11
Solder Listrik 60W Handle Biru

Gambar 2.12
Sedotan Timah (Atraktor)

Gambar 2.13
Multimeter Digital + Probes



Gambar 2.14
Breadboard Prototipe 830 Titik

Gambar 2.15
Tang Potong Sisi Elektronik

Gambar 2.16
Mata Bor Step Drill HSS 4-22mm

Gambar 2.17
Set Obeng Presisi (+ / -)

Gambar 2.18
Kabel DMX XLR 3-Pin 3 Meter

Gambar 2.19
Laptop Acer Swift 3 (Host)

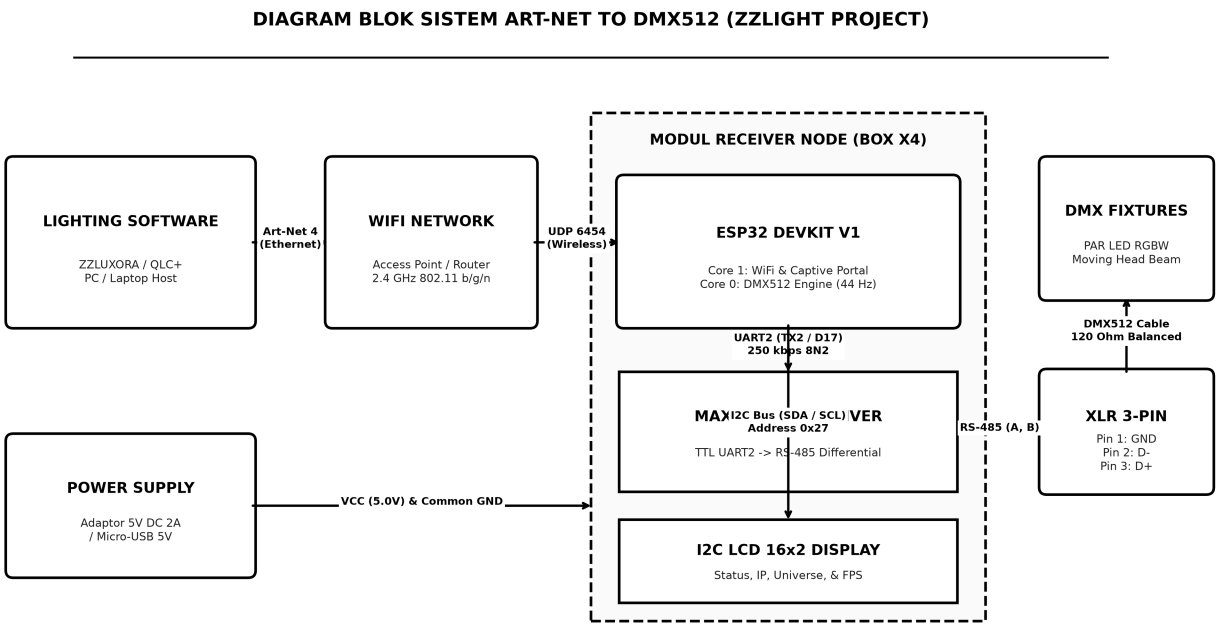
Gambar 2.20
Lampu Panggung PAR LED 54 Mata

Gambar 2.21
Pasta Solder Flux Rosin Paste

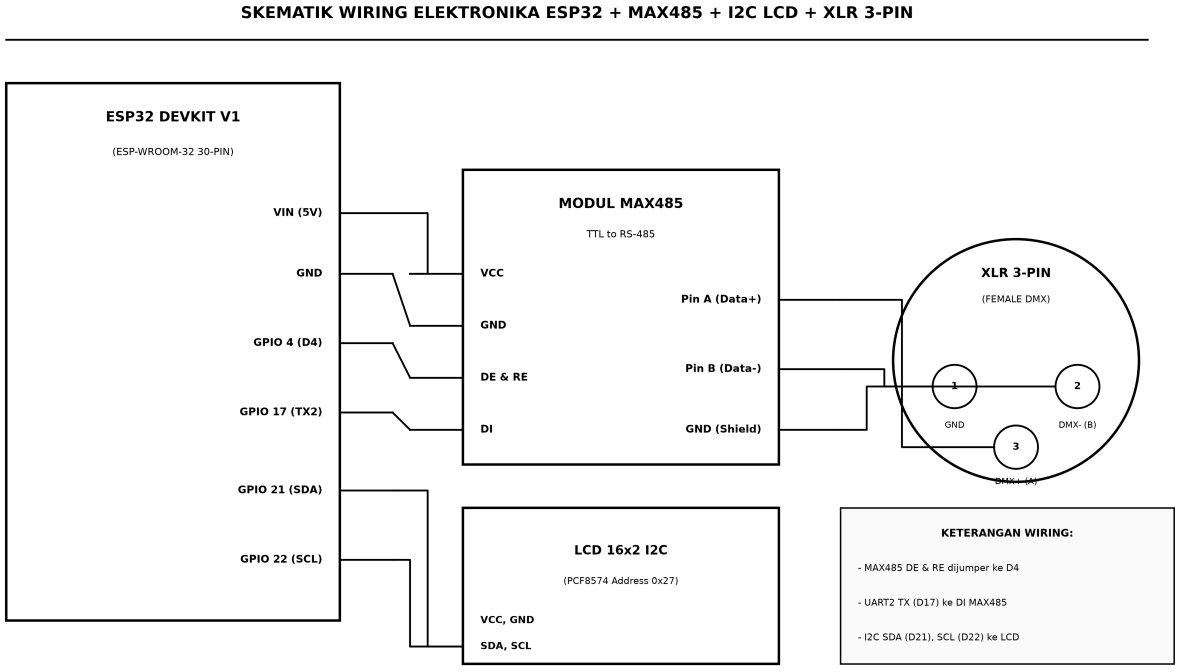
Gambar 2.22
Gunting Craft Elektronik

4. Diagram Skematik & Arsitektur Sistem

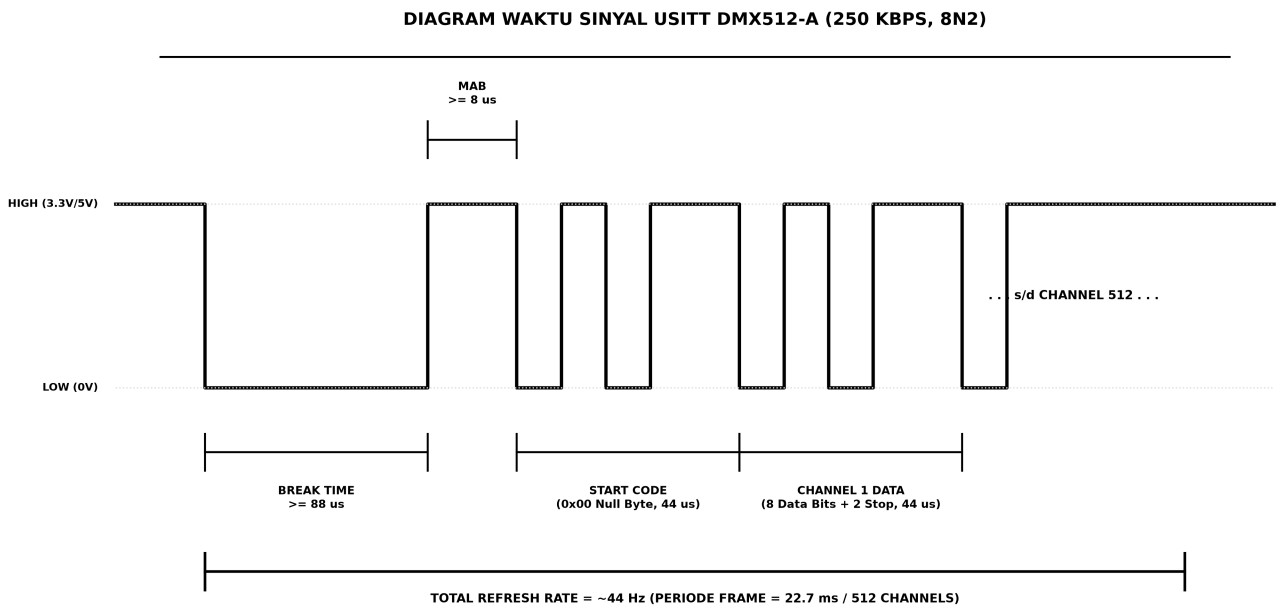
Berikut adalah 5 diagram skematik dan arsitektur sistem (format 300 DPI blueprint) yang digunakan sebagai acuan perancangan:



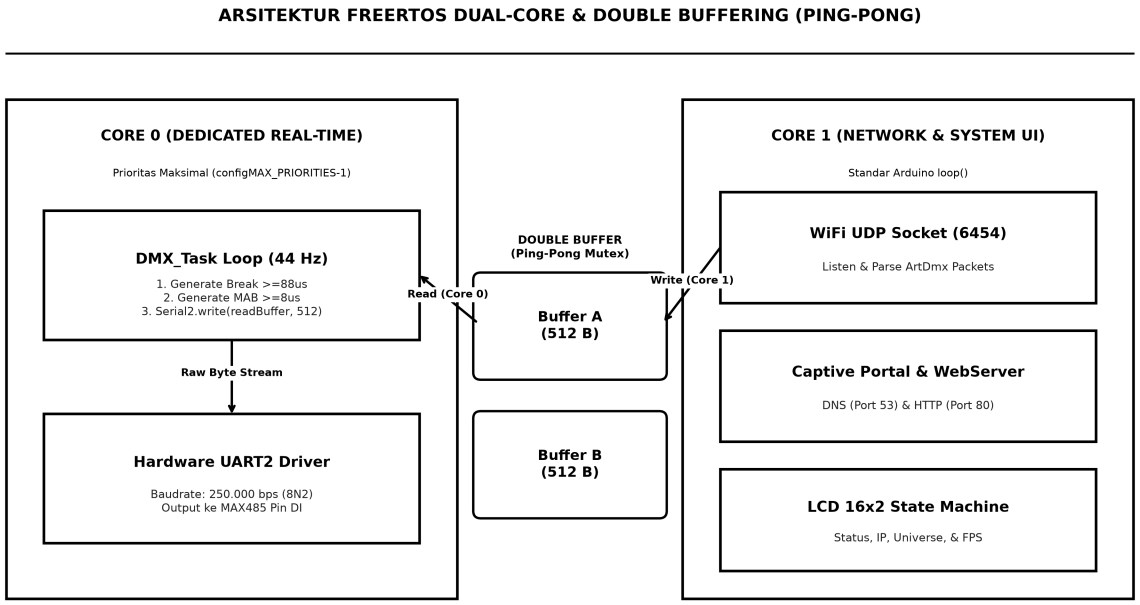
Gambar 3.1: Diagram Blok Alur Komunikasi Sistem Art-Net ke DMX512



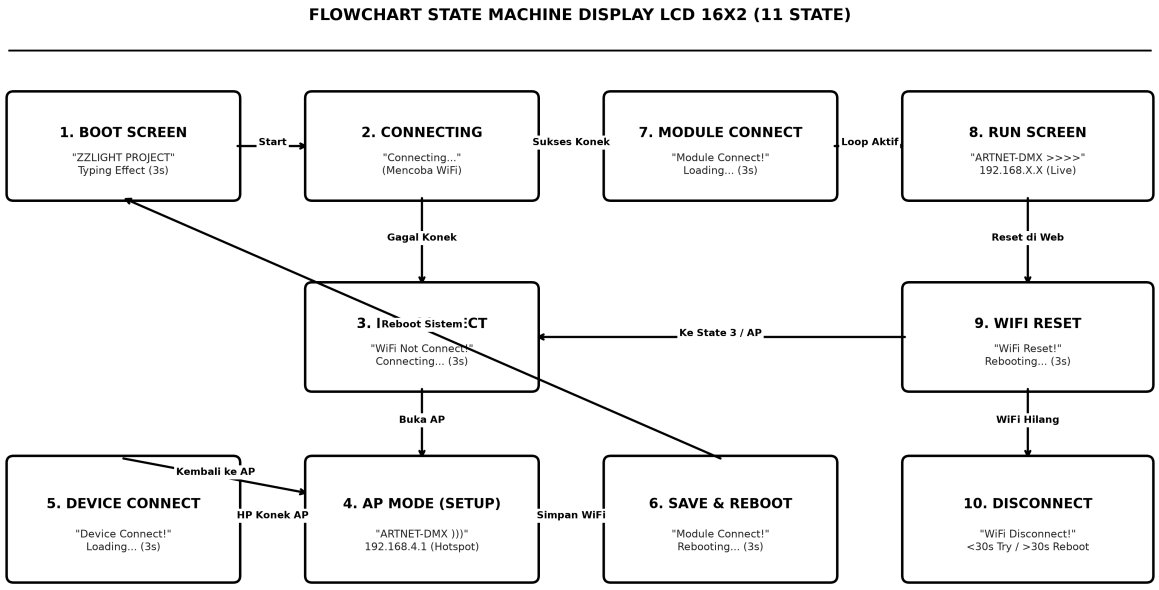
Gambar 3.2: Skematik Perkabelan Detail Pinout ESP32, MAX485, LCD I2C, dan Soket XLR



Gambar 3.3: Diagram Gelombang Waktu Sinyal Standar USITT DMX512-A (250 kbps, 8N2)



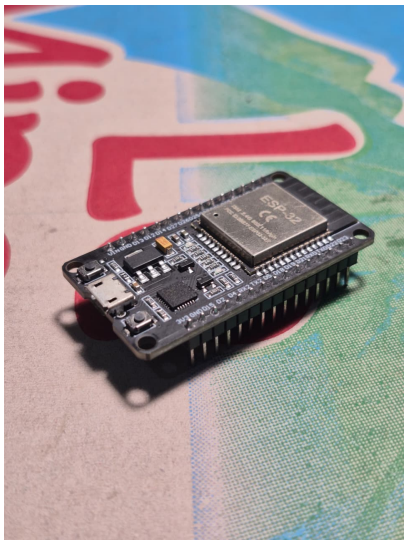
Gambar 3.4: Arsitektur Komputasi Paralel FreeRTOS Dual-Core SMP dan Double Buffering



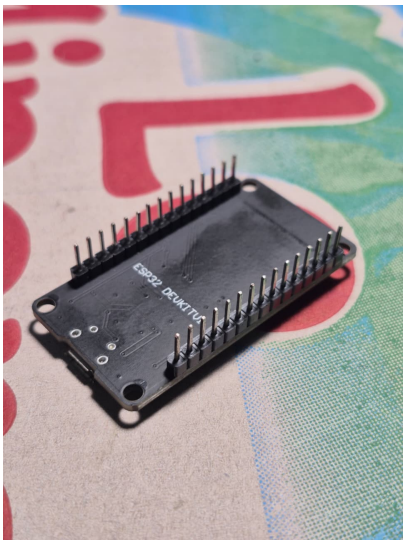
Gambar 3.5: Diagram Alir Mesin Status (11-State Machine) Display LCD 16x2 I2C

5. Dokumentasi Tahapan Perakitan Fisik (Build Log)

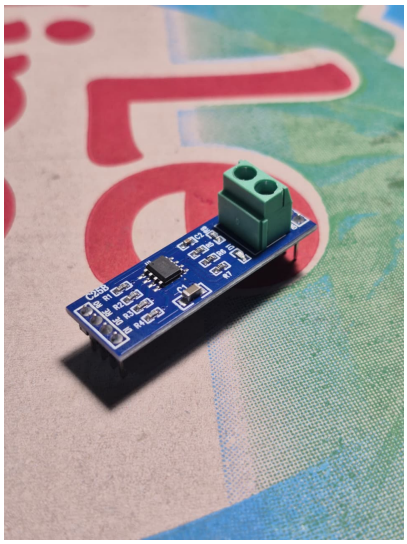
Dokumentasi 30 foto autentik tahapan pengerjaan fisik modul: pengeboran box X4, penyolderan jalur rawboard, perakitan internal, dan pengujian live:



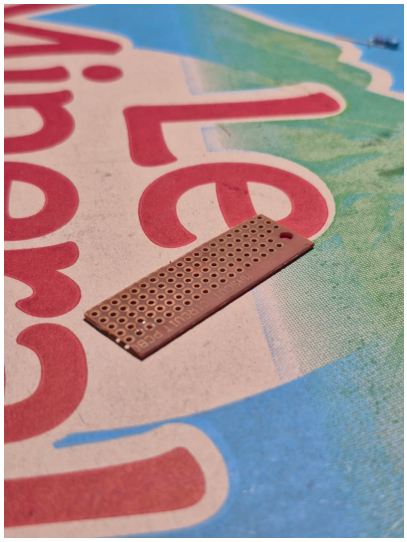
Gambar 4.1
ESP32 DevKit V1 (Tampak Atas)



Gambar 4.2
ESP32 DevKit V1 (Tampak Bawah)

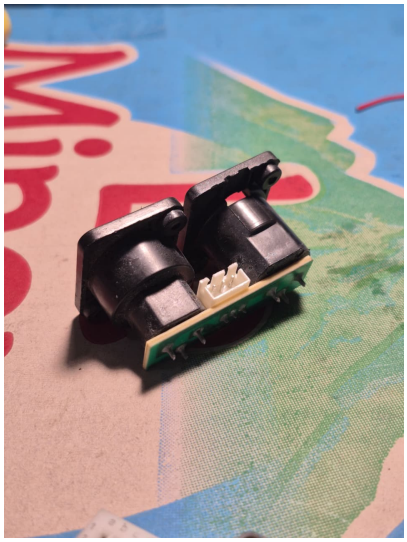


Gambar 4.3
Modul Transceiver MAX485



Gambar 4.4
Papan PCB Matrix (Rawboard)

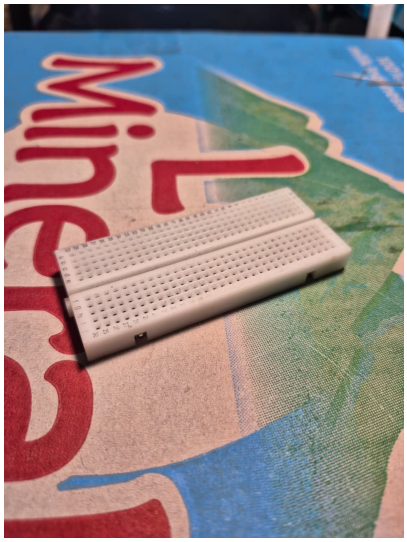
Gambar 4.5
Soket Sasis XLR 3-Pin Depan



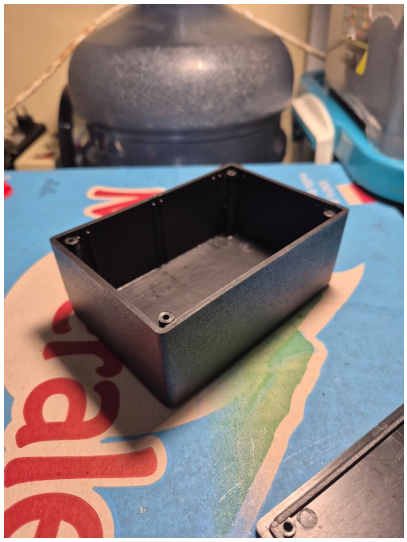
Gambar 4.6
Soket Sasis XLR Terminal Belakang

Gambar 4.7
Box X4 Kondisi Tertutup

Gambar 4.8
Box X4 Dua Sisi Terbuka



Gambar 4.9
Pengujian Prototipe di Breadboard



Gambar 4.10
Penandaan Posisi Lubang Box X4

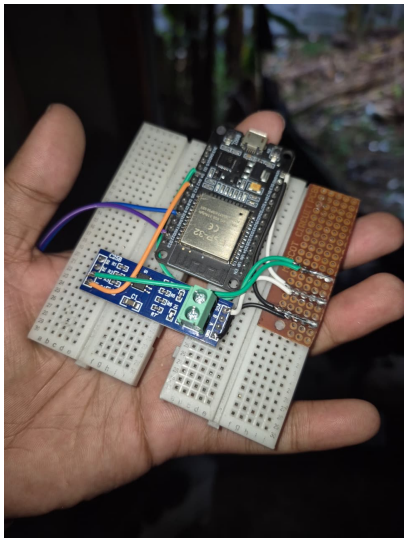
Gambar 4.11
Pengeboran Awal Sasis XLR

Gambar 4.12
Pembesaran Lubang Mata Step Drill

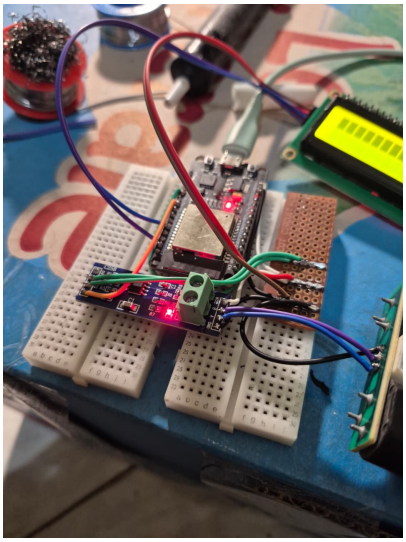
Gambar 4.13
Perapian Lubang Sasis XLR

Gambar 4.14
Pemasangan Soket XLR Depan

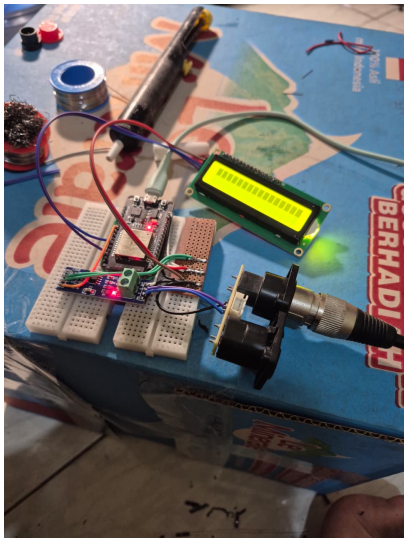
Gambar 4.15
Penguncian Baut Soket XLR



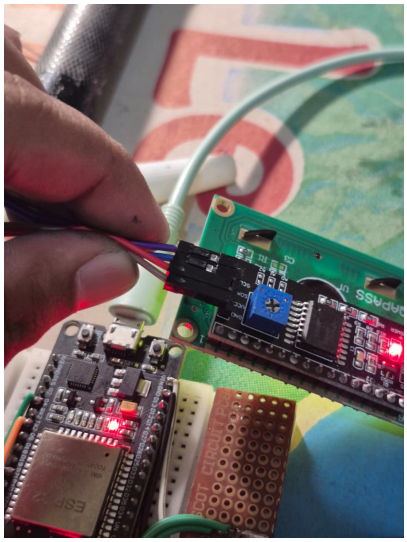
Gambar 4.16
Penyolderan Header Rawboard



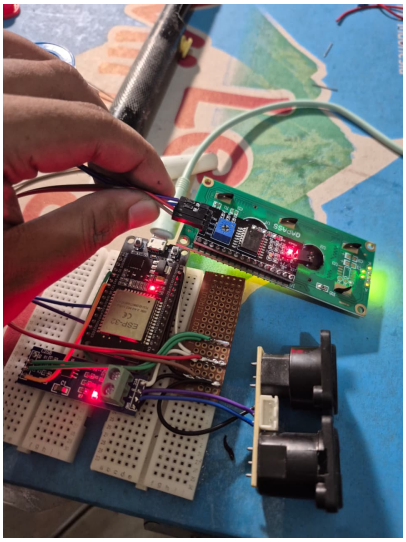
Gambar 4.17
Penempatan Soket Modul ESP32



Gambar 4.18
Perkabelan Jalur Solder Bawah

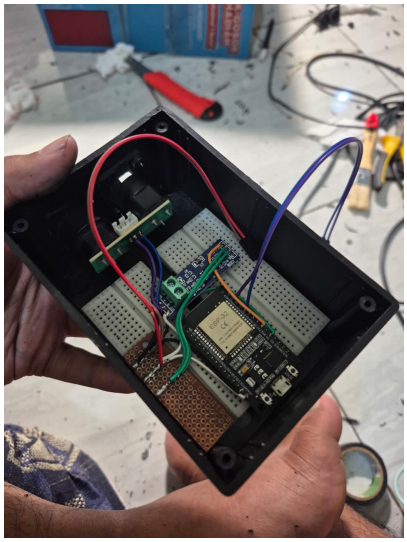


Gambar 4.19
Pengkabelan Jalur I2C LCD 16x2



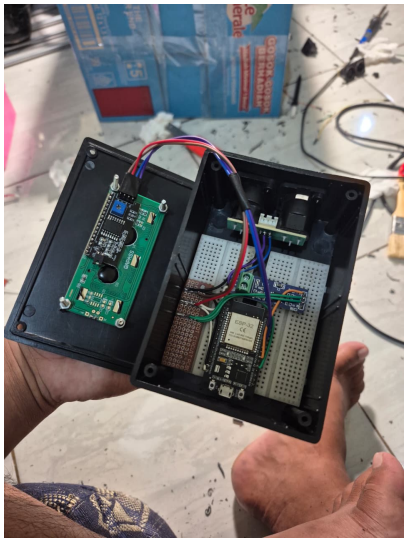
Gambar 4.20
Modul Sirkuit Selesai Disolder

Gambar 4.21
Penempatan Sirkuit ke Box X4

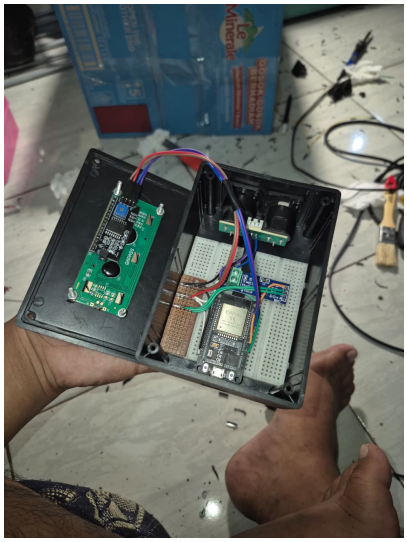


Gambar 4.22
Perakitan Lengkap Dalam Box X4

Gambar 4.23
Pengkabelan Terminal Sasis XLR



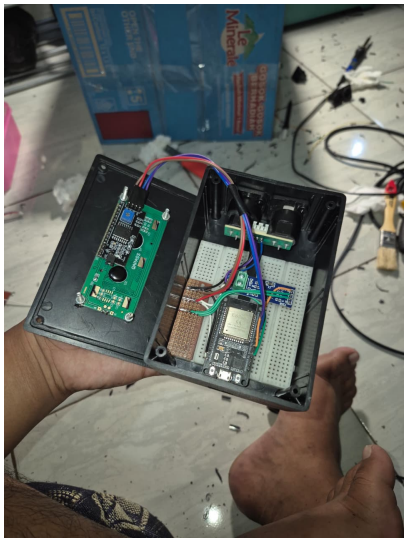
Gambar 4.24
Pemasangan Tutup Atas Box X4



Gambar 4.25
Penguncian Sekrup Enclosure

Gambar 4.26
Produk Modul Jadi Utuh Mandiri

Gambar 4.27
Modul Aktif Menyala di Meja Uji



Gambar 4.28
Tampilan Layar LCD Operasional

Gambar 4.29
Uji Coba Koneksi Lampu PAR LED

Gambar 4.30
Aktuasi Warna Lampu Panggung Live

6. Hasil Pengujian & Roadmap v5.0

Ringkasan Hasil Uji:

• Refresh Rate DMX: 44.1 Hz Deterministik (Zero-Jitter < 0.05 µs).
• Latensi WiFi UDP: 1.8 – 4.2 ms (Jarak 30 meter stabil).
• Fail-Safe Blackout: 10.0 Detik otomatis padam saat sinyal laptop terputus.
• Kompatibilitas Software: Sukses 100% dengan QLC+, grandMA3, Resolume Arena, TouchDesigner, dan ZZLUXORA.

Ide Pengembangan v5.0 (Community Edition):

Bagi siapa pun yang ingin mengembangkan versi berikutnya (v5.0.0), sangat silakan untuk di-fork! Fitur yang disarankan meliputi: protokol sACN E1.31, multi-universe 2-4 port XLR, komunikasi RDM dua arah, update firmware OTA via web, port Ethernet PoE W5500, dan display OLED grafis.